Causal identification in macro I

Version 1 (April 21, 2026)

ECON 31730, Spring 2026
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Semi-structural identification

- If we have a fully-specified structural DSGE model, we can estimate it using MLE/Bayes. Can then use the model to answer all kinds of counterfactual/causal economic questions.
- But DSGE models rely on many tenuous assumptions.
- **Semi-structural identification**: Can we answer interesting structural economic questions if we don't impose a fully-specified DSGE model?
- Examples of questions:
  1. What happens to the economy after a monetary policy shock? (impulse response)
  2. How important are monetary policy shocks as drivers of economic fluctuations generally? (variance decomposition)
  3. Did monetary policy shocks contribute to the recovery from the Great Recession? (historical decomposition)
- Note: This is very different from forecasting!

From DSGE model to SVMA

- Linearized DSGE models can usually be written in the form

$$y_t = c + Hs_t + u_t, \quad s_t = \Theta_1 s_{t-1} + \Theta_\varepsilon \varepsilon_t.$$

- Since we explicitly solve for a non-explosive solution for the states s_t, the polynomial $\Xi(L) \equiv (I_n - \Theta_1 L)^{-1}\Theta_\varepsilon$ exists, and

$$s_t = \Xi(L)\varepsilon_t \quad \Longrightarrow \quad y_t = c + H\Xi(L)\varepsilon_t + u_t.$$

- Defining $\tilde{\varepsilon}_t \equiv (\varepsilon_t', u_t')'$ and $\tilde{\Theta}(L) = (H\Xi(L), I_n)$, we obtain

$$y_t = c + \tilde{\Theta}(L)\tilde{\varepsilon}_t,$$

  where $\tilde{\Theta}(L)$ is $n \times \dim(\tilde{\varepsilon}_t)$.

- The observables y_t are a one-sided moving average in the shocks $\tilde{\varepsilon}_t$ (possibly not invertible).

Running example: permanent income model

- Example: permanent income model (Fernández-Villaverde, Rubio-Ramírez, Sargent & Watson, 2007).
- Exogenous labor income process:

$$x_t = \varepsilon_t \sim WN(0, \sigma_\varepsilon^2).$$

- Consumption function ($R \geq 1$ is gross interest rate):

$$c_t = c_{t-1} + (1 - R^{-1})\varepsilon_t.$$

- Suppose the observable is net savings $s_t \equiv x_t - c_t$. Model implies

$$s_t = s_{t-1} + R^{-1}\varepsilon_t - \varepsilon_{t-1}.$$

- MA(1) model for $y_t \equiv \Delta s_t$.

SVMA model

- From now on we will consider a general **Structural Vector Moving Average (SVMA)** model

$$y_t = \Theta(L)\varepsilon_t, \quad \Theta(L) \equiv \sum_{\ell=0}^{\infty} \Theta_\ell L^\ell,$$

  where I assume mean zero for simplicity.

- We have seen that this is consistent with almost all DSGE models.

- Assume ε_t is n_ε-dimensional mutually uncorrelated white noise:

$$\varepsilon_t \sim WN(0, \operatorname{diag}(\sigma_1^2, \ldots, \sigma_{n_\varepsilon}^2)).$$

  Should think of ε_t as not just structural economic shocks but also possibly measurement error. Mutually uncorrelated because supposed to represent conceptually different economic forces.

- $\{y_t\}$ is observed but $\{\varepsilon_t\}$ is not. Note: Might well have $n_\varepsilon > n$.

SVMA: parameters

$$y_t = \Theta(L)\varepsilon_t, \quad \varepsilon_t \sim WN(0, \mathrm{diag}(\sigma_1^2, \ldots, \sigma_{n_\varepsilon}^2))$$

- Apart from linearity, the SVMA model is a very flexible and unrestrictive model.
- Many parameters (∞-ly many unless we restrict the MA lag length).
- The parameters of the model may not be directly interpretable in terms of fundamental economic parameters: risk aversion, technological constraints, etc.
- But we shall see that several transformations of the model parameters are very useful in counterfactual/causal macroeconomic analysis.

Impulse response function

- We will often be interested in estimating **impulse responses**

$$\Theta_{i,j,\ell} \equiv (\Theta_\ell)_{i,j} = E^*(y_{i,t+\ell} \mid \varepsilon_{j,t} = 1) - \underbrace{E^*(y_{i,t+\ell} \mid \varepsilon_{j,t} = 0)}_{=0}.$$

  Response of y_i at horizon ℓ with respect to one-unit impulse to ε_j.

- **Impulse response function (IRF):**

$$(\Theta_{i,j,0}, \Theta_{i,j,1}, \Theta_{i,j,2}, \dots).$$

  Dynamic profile of how y_i responds over time to shock ε_j.

- Sometimes we look at **relative impulse responses** $\Theta_{i,j,\ell}/\Theta_{1,j,0}$, say. Can be interpreted as the response in y_i at horizon ℓ to a shock ε_j of a magnitude that raises variable y_1 by one unit *on impact*.

Running example: impulse response functions

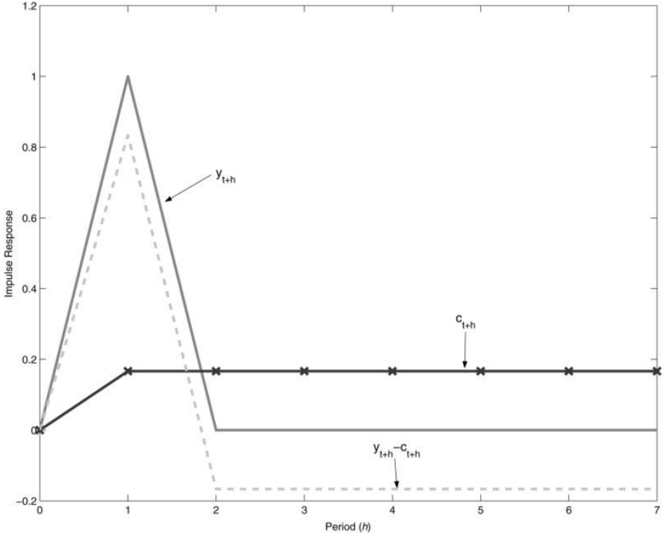

FIGURE 1. IMPULSE RESPONSES OF y_{t+h}, c_{t+h}, AND $y_{t+h} - c_{t+h}$ TO A SHOCK IN w_t

Source: Fernández-Villaverde, Rubio-Ramírez, Sargent & Watson (2007)

SVMA: parameter normalization

- $\Theta(L)$ and $(\sigma_1^2, \ldots, \sigma_{n_\varepsilon}^2)$ are not separately identified:

$$\Theta(L)\varepsilon_t \overset{d}{=} \tilde{\Theta}(L)\tilde{\varepsilon}_t,$$

$$\text{where} \quad \tilde{\Theta}(L) \equiv \textstyle\sum_{\ell=0}^{\infty} \Theta_\ell \operatorname{diag}(\sigma_1, \ldots, \sigma_{n_\varepsilon}) L^\ell, \quad \tilde{\varepsilon}_t \sim WN(0, I_{n_\varepsilon}).$$

- We need to *normalize* some of the parameters. Common alternatives:

  1. **Unit standard deviation:** $\sigma_j = 1$ for all j. One-unit shock to ε_j is a one standard deviation shock.

  2. **Unit effect:** $\Theta_{j,j,0} = 1$ for all j. Then $\Theta_{i,j,\ell}$ is the response of y_i at horizon ℓ to a shock ε_j of a magnitude (namely σ_j) that raises y_j by one unit *on impact*.

- Going forward, I will proceed with the first option above, so $\sigma_j = 1$ henceforth. Beware: Some papers use second normalization.

Variance decomposition

- Impulse responses are useful for quantifying the effects of shocks, but we also want to quantify how *important* shocks are.
- SVMA model:
$$y_{i,t} = \sum_{j=1}^{n_\varepsilon} \sum_{\ell=0}^{\infty} \Theta_{i,j,\ell}\varepsilon_{j,t-\ell}.$$
- The share of the overall variance of $y_{i,t}$ that is due to $\{\varepsilon_{j,t}\}$ is
$$VD_{i,j} \equiv 1 - \frac{\mathrm{Var}^*(y_{i,t} \mid \{\varepsilon_{j,\tau}\}_{-\infty<\tau\leq t})}{\mathrm{Var}(y_{i,t})} = \frac{\mathrm{Var}(\sum_{\ell=0}^{\infty} \Theta_{i,j,\ell}\varepsilon_{j,t-\ell})}{\mathrm{Var}(y_{i,t})}$$
$$= \frac{\sum_{\ell=0}^{\infty} \Theta_{i,j,\ell}^2}{\sum_{\tilde{j}=1}^{n_\varepsilon} \sum_{\ell=0}^{\infty} \Theta_{i,\tilde{j},\ell}^2} \in [0,1].$$
- Note: $\sum_{j=1}^{n_\varepsilon} VD_{i,j} = 1$ for all i.
- Could also define a frequency-specific version of this concept (e.g., only business cycle frequencies), see Plagborg-Moller & Wolf (2022).

Forecast variance decomposition

- We might also like to know how important a shock ε_j is for forecasting variable y_i at horizon ℓ.

- Hence, we seek to decompose variance of

$$y_{i,t+\ell} - E^*(y_{i,t+\ell} \mid \{\varepsilon_\tau\}_{-\infty<\tau\leq t}) = \sum_{j=1}^{n_\varepsilon} \sum_{h=0}^{\ell-1} \Theta_{i,j,h}\varepsilon_{j,t+\ell-h}.$$

- **Forecast (error) variance decomposition (FVD/FEVD):**

$$\begin{aligned} FVD_{i,j,\ell} &\equiv 1 - \frac{\mathrm{Var}^*(y_{i,t+\ell} \mid \{\varepsilon_{j,\tau}\}_{t<\tau\leq t+\ell}, \{\varepsilon_\tau\}_{-\infty<\tau\leq t})}{\mathrm{Var}^*(y_{i,t+\ell} \mid \{\varepsilon_\tau\}_{-\infty<\tau\leq t})} \\ &= \frac{\sum_{h=0}^{\ell-1} \Theta_{i,j,h}^2}{\sum_{\tilde{j}=1}^{n_\varepsilon} \sum_{h=0}^{\ell-1} \Theta_{i,\tilde{j},h}^2} \end{aligned}$$

  Percentage reduction in forecast variance that would occur if we knew the future realizations of ε_j.

- Note: $\sum_{j=1}^{n_\varepsilon} FVD_{i,j,\ell} = 1$ for all i, ℓ.

Running example: variance decompositions

- Permanent income model for change in savings (now unit-standard-deviation normalization):

$$y_t \equiv \Delta s_t = R^{-1}\sigma_\varepsilon \varepsilon_t - \sigma_\varepsilon \varepsilon_{t-1}.$$

- Both *VD* and *FVD* equal 1, since there's only one shock.
- But if y_t is additionally affected by additive measurement error, this would no longer be the case.

Historical decomposition

- Finally, we may be interested in learning about what part of the fluctuations in y_i are due to ε_j, at a *particular* point in time (rather than on average over time).
- **Historical decomposition:**

$$HD_{i,j,t} \equiv E^*(y_{i,t} \mid \{\varepsilon_{j,\tau}\}_{-\infty<\tau\leq t}) = \textstyle\sum_{\ell=0}^{\infty} \Theta_{i,j,\ell}\varepsilon_{j,t-\ell}.$$

- This is a random variable and not a parameter. We will still loosely talk about "identifying" it in the following.
- $(y_{i,t} - HD_{i,j,t})$ is what $y_{i,t}$ *would have* looked like if there had been no ε_j shocks at all.
- Note: $\sum_{j=1}^{n_\varepsilon} HD_{i,j,t} = y_{i,t}$.

SVMA: identification

- If we knew the SVMA parameters $\Theta(L)$, we could just compute all these objects of interest. Can we consistently estimate them?
- Answer: No, because they are not identified w/o further as'ns.
- If data $\{y_t\}$ is purely non-deterministic (and zero mean), Wold decomposition theorem says that there exists *some* representation

$$y_t = \Psi(L)\eta_t = \tilde{\Psi}(L)\tilde{\eta}_t$$

with $\dim(\eta_t) = \dim(\tilde{\eta}_t) = n$, where

$$\eta_t \equiv y_t - E^*(y_t \mid \{y_\tau\}_{-\infty<\tau<t}) \sim WN(0, \Sigma),$$

$$\tilde{\eta}_t \equiv \Sigma^{-1/2}\eta_t \sim WN(0, I_n), \quad \tilde{\Psi}(L) \equiv \Psi(L)\Sigma^{1/2}, \quad \Sigma = \Sigma^{1/2}\Sigma^{1/2\prime}.$$

- But in fact there exist many more SVMA representations!

Example: identification of MA(1)

- Consider first the univariate MA(1) model

$$y_t = \Theta_0 \varepsilon_t + \Theta_1 \varepsilon_{t-1}, \quad \varepsilon_t \sim WN(0,1).$$

  (For example: permanent income model.)

- Exercise: Show that, when $\Theta_0 \neq 0$, the autocovariance function (ACF) of the above process coincides with that of the process

$$y_t = \tilde{\Theta}_0 \tilde{\varepsilon}_t + \tilde{\Theta}_1 \tilde{\varepsilon}_{t-1}, \quad \tilde{\varepsilon}_t \sim WN(0,1),$$

$$(\tilde{\Theta}_0, \tilde{\Theta}_1) = \pm(\Theta_1, \Theta_0).$$

- If Θ_0 is quite different from Θ_1, the impulse responses in the alternative representation are substantially different from the true impulse responses. But observationally equivalent.

Example: identification of MA(2)

- Now consider the univariate MA(2) model

$$y_t = \varepsilon_t - \tfrac{1}{6}\varepsilon_{t-1} - \tfrac{1}{6}\varepsilon_{t-2}, \quad \varepsilon_t \sim WN(0,1). \qquad (\dagger)$$

- Note that

$$\Theta(L) \equiv 1 - \tfrac{1}{6}L - \tfrac{1}{6}L^2 = (1 - \tfrac{1}{2}L)(1 + \tfrac{1}{3}L),$$

so the roots of $\Theta(L)$ (2 and -3) are outside the unit circle. Hence, ($\dagger$) is the Wold representation of $\{y_t\}$.

- The spectral density of the data $\{y_t\}$ is

$$s_y(\omega) = \tfrac{1}{2\pi}|\Theta(e^{-i\omega})|^2.$$

Example: identification of MA(2) (cont.)

$$y_t = \Theta(L)\varepsilon_t, \quad \Theta(L) = (1 - \tfrac{1}{2}L)(1 + \tfrac{1}{3}L), \quad \varepsilon_t \sim WN(0,1)$$

- Now define

  $$\tilde{\Theta}(L) \equiv (L - \tfrac{1}{2})(1 - \tfrac{1}{2}L)^{-1}\Theta(L) = (L - \tfrac{1}{2})(1 + \tfrac{1}{3}L) = -\tfrac{1}{2} + \tfrac{5}{6}L + \tfrac{1}{3}L^2.$$

  This lag polynomial has roots $\frac{1}{2}$ and -3. We "flipped" one of the roots of $\Theta(L)$.

- Spectral density of $\{\tilde{y}_t \equiv \tilde{\Theta}(L)\varepsilon_t\}$:

  $$s_{\tilde{y}}(\omega) = \left| \frac{e^{-i\omega} - \frac{1}{2}}{1 - \frac{1}{2}e^{-i\omega}} \right|^2 s_y(\omega) = s_y(\omega).$$

- Exercise: Show $|\frac{e^{-i\omega}-a}{1-ae^{-i\omega}}| = 1$ for all $\omega \in [0, 2\pi]$, $a \in \mathbb{C}$ s.t. $|a| < 1$.

- So $\{y_t\}$ has the exact same spectral density (second moments) as $\{\tilde{y}_t\}$, even though the MA parameters are different!

SVMA: identification (cont.)

- Lesson: We can generate MA processes with observationally equivalent second moments by **flipping roots** of the MA polynomial.
- Multivariate version of root flipping: Lippi & Reichlin (1994).
- There is another source of lack of identification if we have more than one shock. Consider again the multivariate SVMA

$$y_t = \Theta(L)\varepsilon_t, \quad \varepsilon_t \sim WN(0, I_{n_\varepsilon}). \tag{‡}$$

- The $n \times n_\varepsilon$ SVMA polynomials $\Theta(L)$ and $\tilde{\Theta}(L) \equiv \Theta(L)Q$ generate the same spectral density if $QQ' = I_n$:

$$\Theta(e^{-i\omega})\Theta(e^{-i\omega})^* = \Theta(e^{-i\omega})QQ'\Theta(e^{-i\omega})^* = \tilde{\Theta}(e^{-i\omega})\tilde{\Theta}(e^{-i\omega})^*.$$

- Such a matrix Q is called an **orthogonal matrix** or **rotation matrix**.

SVMA: identification (cont.)

- The spectral density (i.e., ACF) of the data is all we have to identify the parameters with. Model only restricts first and second moments.
- Conclusion: The basic SVMA model (‡) is grossly under-identified.
- We need to make further assumptions in order to learn something useful from the data about our objects of interest.
- Makes sense: We haven't even attached any economic interpretation to the elements of ε_t yet, so how could we hope to learn about the effects/importance of different types of shocks?

Running example: identification

$$y_t = \Delta s_t = \underbrace{R^{-1}\sigma_\varepsilon}_{\Theta_0}\varepsilon_t + \underbrace{(-\sigma_\varepsilon)}_{\Theta_1}\varepsilon_{t-1}, \quad \varepsilon_t \sim WN(0,1)$$

- This is an MA(1) which is not generally identified, as we saw.
- However, economic theory gives the restriction $R \geq 1$, so $|\Theta_0| \leq |\Theta_1|$. Also, $\Theta_0 > 0$.
- Exercise: Show that these two restrictions suffice to point-identify Θ_0 and Θ_1 from $\text{Var}(y_t)$ and $\text{Cov}(y_t, y_{t-1})$.
- Do we always need a fully-specified DSGE model to guide identification in SVMA models? No, as we shall now see.

Invertibility assumption

- Until recently, almost the entire literature on semi-structural identification has chosen to ameliorate the identification problem by assuming that the shocks are **invertible** (a.k.a. **fundamental**):

$$\varepsilon_t \in \overline{sp}(y_\tau, -\infty < \tau \leq t).$$

- Requires that we can recover shocks as (linear) functions of *current* and *past* (but not future) data.

- Invertibility implies

$$\overline{sp}(\varepsilon_\tau, -\infty < \tau \leq t) = \overline{sp}(y_\tau, -\infty < \tau \leq t).$$

  I.e., an econometrician observing $\{y_\tau\}_{\tau \leq t}$ can forecast as accurately as an economic agent who observes the underlying shocks $\{\varepsilon_\tau\}_{\tau \leq t}$.

- Highly restrictive! Everything that is a surprise to the econometrician must also be a surprise to agents in the (unknown) underlying model.

Invertibility assumption (cont.)

$$\varepsilon_t \in \overline{sp}(y_\tau, -\infty < \tau \leq t).$$

- The invertibility assumption depends on the set of observables y_t.
- The more relevant (!) data we observe, the more plausible it is to assume invertibility.
- We certainly need $n_\varepsilon \leq n$ (necessary, not sufficient).
- But simply adding data blindly won't do, as the extra series could be contaminated by additional shocks.
- In fact, many DSGE models have non-invertible SVMA representations for standard sets of observables.

Running example: invertibility

$$\Delta c_t = (1 - R^{-1})\sigma_\varepsilon \varepsilon_t, \quad x_t = \sigma_\varepsilon \varepsilon_t,$$
$$y_t \equiv \Delta(x_t - c_t) = \Theta(L)\varepsilon_t, \quad \Theta(L) \equiv \sigma_\varepsilon(R^{-1} - L).$$

- If we observed Δc_t or x_t, then ε_t would clearly be invertible. What if we only observe y_t?
- Since the root of $\Theta(z)$ ($z = R^{-1}$) lies inside the unit circle, $\Theta(L)$ does not have a one-sided inverse. Thus, cannot represent ε_t as function of just current and lagged y_t.
- Can also see this by repeated substitution:

$$\sigma_\varepsilon \varepsilon_t = R(y_t + \sigma_\varepsilon \varepsilon_{t-1}) = Ry_t + R^2(y_{t-1} + \sigma_\varepsilon \varepsilon_{t-2})$$
$$= \cdots = \textstyle\sum_{\ell=1}^K R^\ell y_{t-\ell+1} + R^K \sigma_\varepsilon \varepsilon_{t-K}.$$

Doesn't converge in mean-square as $K \to \infty$, since $R \geq 1$.

Running example: invertibility (cont.)

- So if we only observe net savings, the shock ε_t is not invertible: We cannot recover ε_t linearly from the infinite history $\{y_\tau\}_{\tau \le t}$.
- Note: We can recover ε_t from *future* values $\{y_\tau\}_{\tau > t}$. More on this later in the course.
- Exercise: Verify by direct substitution that

$$\sigma_\varepsilon \varepsilon_t = -\sum_{\ell=1}^{K} R^{1-\ell} y_{t+\ell} + R^{-K} \sigma_\varepsilon \varepsilon_{t+K} \overset{ms}{\to} -\sum_{\ell=1}^{\infty} R^{1-\ell} y_{t+\ell},$$

as $K \to \infty$.

Invertibility and Wold innovations

- Define Wold innovation $\eta_t \equiv y_t - E^*(y_t \mid \{y_\tau\}_{-\infty<\tau<t})$.
- Since (from Wold theorem)

$$\overline{sp}(y_\tau, -\infty < \tau \leq t) = \overline{sp}(\eta_\tau, -\infty < \tau \leq t),$$

  the invertibility assumption can equivalently be stated as

$$\varepsilon_t \in \overline{sp}(\eta_\tau, -\infty < \tau \leq t).$$

- But ε_t is uncorrelated with $\{y_\tau\}_{\tau<t}$, and thus uncorrelated with $\{\eta_\tau\}_{\tau<t}$. Moreover, $\{\eta_t\}$ is white noise.
- Hence, invertibility means that $\varepsilon_t \in sp(\eta_t)$, or in other words, that there exists a matrix $G \in \mathbb{R}^{n_\varepsilon \times n}$ s.t.

$$\varepsilon_t = G\eta_t.$$

  The structural shocks are linear combinations of the *current* reduced-form forecast errors. Not true under non-invertibility!

Running example: Wold innovation

- From the MA(1) identification example earlier, we know that the Wold decomposition (invertible MA(1) representation) of y_t is obtained by swapping Θ_0 and Θ_1, i.e.,

$$y_t = -\sigma_\varepsilon \tilde{\varepsilon}_t + R^{-1}\sigma_\varepsilon \tilde{\varepsilon}_{t-1} = \eta_t - R^{-1}\eta_{t-1}, \quad \eta_t \equiv -\sigma_\varepsilon \tilde{\varepsilon}_t \sim WN(0, \sigma_\varepsilon^2).$$

- By definition, the Wold innovation (forecast error) is invertible:

$$\eta_t = (1 - R^{-1}L)^{-1} y_t = \textstyle\sum_{\ell=0}^{\infty} R^{-\ell} y_{t-\ell}.$$

- Plug in for y_t to get an expression for η_t in terms of the true shocks:

$$\begin{aligned}\eta_t &= \textstyle\sum_{\ell=0}^{\infty} R^{-\ell}\sigma_\varepsilon (R^{-1}\varepsilon_{t-\ell} - \varepsilon_{t-\ell-1}) \\ &= R^{-1}\sigma_\varepsilon \left\{\varepsilon_t + (1 - R^2)\textstyle\sum_{\ell=1}^{\infty} R^{-\ell}\varepsilon_{t-\ell}\right\}.\end{aligned}$$

  Exercise: Verify the last expression.

- Clearly, ε_t cannot be obtained as a linear function of just η_t.

From SVMA to SVAR under invertibility

- Back to the general SVMA model:

$$y_t = \Theta(L)\varepsilon_t, \quad \varepsilon_t \sim WN(0, I_{n_\varepsilon}).$$

- Like much of the older semi-structural literature, we will for now impose the invertibility assumption. (We will later discuss methods that relax this.)
- In particular, $\Theta(L)$ is assumed to be square $n \times n$.
- Under invertibility, the roots of $\Theta(L)$ cannot be *inside* the unit circle. But they could still potentially be *on* the unit circle.
- Assume in addition to invertibility that the spectral density matrix $s_y(\omega)$ is non-singular for all ω.
- Then the one-sided inverse $\tilde{A}(L) = \sum_{\ell=0}^{\infty} \tilde{A}_\ell L^\ell \equiv \Theta(L)^{-1}$ exists.

From SVMA to SVAR under invertibility (cont.)

$$y_t = \Theta(L)\varepsilon_t \quad \Longrightarrow \quad \varepsilon_t = \Theta(L)^{-1}y_t = \tilde{A}(L)y_t = \textstyle\sum_{\ell=0}^{\infty} \tilde{A}_\ell y_{t-\ell}$$

- We obtain the **Structural VAR (SVAR)** model

$$A(L)y_t = H\varepsilon_t, \quad A(L) = I_n - \textstyle\sum_{\ell=1}^{\infty} A_\ell L^\ell,$$

  where $H = \tilde{A}_0^{-1}$ and $A(L) = \tilde{A}_0^{-1}\tilde{A}(L)$.

- An SVAR model is an SVMA model that assumes invertibility (and non-singular spectrum). Nothing more, nothing less.

SVAR: identification

$$A(L)y_t = H\varepsilon_t, \quad A(L) = I_n - \sum_{\ell=1}^{\infty} A_\ell L^\ell$$

- Even given the invertibility assumption needed to arrive at an SVAR model, there is still an identification problem.
- The data allows us to identify the VAR coef's $A(L)$ and forecast errors

$$\eta_t \equiv y_t - E^*(y_t \mid \{y_\tau\}_{-\infty<\tau<t}) = A(L)y_t = H\varepsilon_t,$$

which are white noise. But the only information available for identifying H is the var-cov matrix

$$\Sigma \equiv \mathrm{Var}(\eta_t) = HH'.$$

$\frac{n(n+1)}{2}$ equations (due to symmetry), but n^2 unknowns.

SVAR: identification (cont.)

$$A(L)y_t = \eta_t, \quad \eta_t = H\varepsilon_t, \quad \Sigma \equiv \mathrm{Var}(\eta_t) = HH'$$

- Let C denote the **Cholesky decomposition** of Σ. This is the unique *lower triangular* $n \times n$ matrix with positive diagonal such that

$$\Sigma = CC'.$$

  (Matlab command: `chol`, but beware transposed definition.)

- Exercise: Prove there exists an $n \times n$ orthogonal matrix Q s.t.

$$H = CQ.$$

- Thus, the data identifies $A(L)$ and C, but is completely silent about the **rotation matrix** Q.

- The invertibility assumption "got rid of" the root-flipping issue in SVMA identification, but the rotational indeterminacy remains.

SVAR: identification (cont.)

$$A(L)y_t = \eta_t, \quad \eta_t = H\varepsilon_t, \quad H = CQ, \quad C = \text{chol}(\Sigma)$$

- If we knew Q, we could identify the impact impulse responses $H = CQ$ and the shocks $\varepsilon_t = H^{-1}\eta_t$.
- We could then identify the structural impulse responses $\Theta(L)$ in the SVMA representation as follows:
  1. Compute the *reduced-form* impulse responses (i.e., to the innovations η_t)
  $$B(L) = I_n + \textstyle\sum_{\ell=1}^{\infty} B_\ell L^\ell \equiv A(L)^{-1}.$$
  As usual, these can be computed recursively from the VAR coef's.
  2. Compute $\Theta(L) = B(L)H$.
- Note: If we only care about the IRFs and FVDs for the first shock ε_1, we only need to identify the first column of H, i.e., the first column of Q. This is called **single-shock identification**.
- Next: Which assumptions can help pin down Q?

SVAR: short-run restrictions

- Following the seminal paper by Sims (1980), many papers have imposed a **recursive/Cholesky** identification scheme: $Q = I_n$.

- Then $H = C = \text{chol}(\Sigma)$, and thus

$$A(L)y_t = \begin{pmatrix} H_{11}\varepsilon_{1,t} \\ H_{21}\varepsilon_{1,t} + H_{22}\varepsilon_{2,t} \\ \vdots \\ H_{n1}\varepsilon_{1,t} + H_{n2}\varepsilon_{2,t} + \cdots + H_{nn}\varepsilon_{n,t} \end{pmatrix}.$$

- First variable y_1 responds on impact to ε_1 but not other shocks (only with a lag). y_2 responds on impact to first two shocks, etc.

- Typically justified by appealing to timing assumptions. "GDP growth responds to monetary policy with a lag due to inertia."

- Intuitively simple, but seldom justified by standard DSGE models.

SVAR: estimation and inference

- We have only talked about (population) *identification* so far.
- To estimate parameters, we approximate the VAR(∞) with a finite-order VAR(p).
- Then apply the same transformations to the estimated VAR parameters as in the population formulas on previous slides.
- Gives us estimates of IRFs and FVDs. For the latter, get denominator from standard VAR forecasting formula.
- Inference: delta method (e.g., in Stata), bootstrap, or Bayes.
- Historical decomposition: plug in estimated shocks $\hat{\varepsilon}_t \equiv \hat{H}^{-1}\hat{\eta}_t$, where $\hat{\eta}_t$ are VAR forecast errors.

Application: Christiano, Eichenbaum & Evans (2005)

- Estimate a SVAR using a recursive identification scheme.

$$A(L)\begin{pmatrix} y_t^{(1)} \\ r_t \\ y_t^{(2)} \end{pmatrix} = \begin{pmatrix} H_{11}\varepsilon_{1,t} \\ H_{21}\varepsilon_{1,t} + H_{22}\varepsilon_{2,t} \\ \vdots \\ H_{n1}\varepsilon_{1,t} + \cdots + H_{nn}\varepsilon_{n,t} \end{pmatrix},$$

  where r_t is the federal funds rate.

- Let $m = \dim(y_t^{(1)})$. Monetary policy shock $= \varepsilon_{m+1}$.
- "Slow-moving" variables $y_t^{(1)}$ do not respond to monetary policy shock on impact.
- "Fast-moving var's" $y_t^{(2)}$ may respond to monetary shock on impact, but r_t does not respond to the shocks that impact $y_t^{(2)}$ and not $y_t^{(1)}$.
- Exercise: Show that monetary policy shock is given by a "Taylor rule residual" $\varepsilon_{m+1,t} \propto r_t - E^*(r_t \mid y_t^{(1)}, \{y_\tau\}_{-\infty<\tau<t})$.

Application: Christiano, Eichenbaum & Evans (2005)

- $y_t^{(1)}$: real GDP, real cons'n, GDP deflator, real investment, real wage, labor productivity.
- $y_t^{(2)}$: real profits, growth rate of M2.
- Terminology: $y_t^{(1)}$ is *ordered before* r_t, while $y_t^{(2)}$ is *ordered after* r_t.

Application: Christiano, Eichenbaum & Evans (2005)

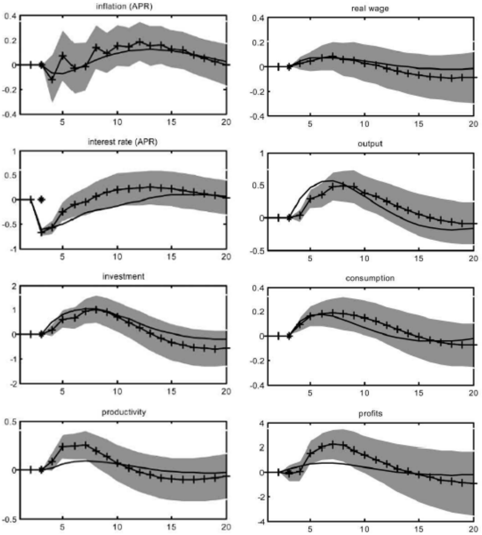

FIG. 1.—Model- and VAR-based impulse responses. Solid lines are benchmark model impulse responses; solid lines with plus signs are VAR-based impulse responses. Grey areas are 95 percent confidence intervals about VAR-based estimates. Units on the horizontal axis are quarters. An asterisk indicates the period of policy shock. The vertical axis units are deviations from the unshocked path. Inflation, money growth, and the interest rate are given in annualized percentage points (APR); other variables are given in percentages.

Source: Christiano, Eichenbaum & Evans (2005)

Note: Some impact impulse responses are zero by construction.

Application: Christiano, Eichenbaum & Evans (2005)

TABLE 1
PERCENTAGE VARIANCE DUE TO MONETARY POLICY SHOCKS

	4 Quarters Ahead	8 Quarters Ahead	20 Quarters Ahead
Output	15	38	27
	(4,26)	(15,48)	(9,35)
Inflation	1	4	7
	(0,8)	(1,11)	(3,18)
Consumption	14	21	14
	(4,26)	(5,37)	(4,26)
Investment	10	26	23
	(2,21)	(7,39)	(6,32)
Real wage	2	2	4
	(0,8)	(0,14)	(0,15)
Productivity	15	14	10
	(3,25)	(3,26)	(3,20)
Federal funds rate	32	19	18
	(18,44)	(8,27)	(5,27)
M2 growth	19	19	19
	(8,29)	(8,26)	(8,24)
Real profits	13	18	7
	(5,25)	(6,31)	(2,20)

NOTE.—Numbers in parentheses are the boundaries of the associated 95 percent confidence interval.

Source: Christiano, Eichenbaum & Evans (2005)

SVAR: long-run restrictions

- Suppose the data y_t equals first diff's of some other data: $y_t = \Delta w_t$. Consider case $n = n_\varepsilon = 2$ (Blanchard & Quah, 1989).
- Suppose we believe that the *long-run level* of the variable w_1 is only driven by ε_1 and not by ε_2 (e.g., real output level only driven by supply shocks in long run, not by demand shocks):

$$0 = \lim_{\ell \to \infty} E^*(w_{1,t+\ell} - w_{1,t-1} \mid \varepsilon_{2,t} = 1) = \sum_{\ell=0}^{\infty} E^*(y_{1,t+\ell} \mid \varepsilon_{2,t} = 1).$$

- That is, the *cumulative* IRF of y_1 should be zero for shock 2:

$$0 = \sum_{\ell=0}^{\infty} \Theta_{1,2,\ell} = (\Theta(1))_{1,2} = (B(1)CQ)_{1,2} = B_{1\bullet}(1)CQ_{\bullet 2}.$$

- We can solve for the rotation matrix $Q = (Q_{\bullet 1}, Q_{\bullet 2})$ from the above restriction on $Q_{\bullet 2}$ and from $QQ' = I_2$.
- Note: $B(1) = A(1)^{-1} = (I_n - \sum_{\ell=1}^{\infty} A_\ell)^{-1}$.

Partial invertibility

- So far, when discussing SVAR identification we have been working under the assumption that *all* shocks are invertible, in the sense

$$\varepsilon_t = (\varepsilon_{1,t}, \ldots, \varepsilon_{n_\varepsilon,t})' \in \overline{sp}(y_\tau, -\infty < \tau \leq t).$$

- However, it turns out that if we want to estimate IRFs with respect to a single shock (e.g., the first one ε_1) with SVARs, then we only need to assume **partial invertibility**:

$$\varepsilon_{1,t} \in \overline{sp}(y_\tau, -\infty < \tau \leq t).$$

  Other shocks may be non-invertible: $\varepsilon_{j,t} \notin \overline{sp}(y_\tau, \tau \leq t)$, $j > 1$.

- Still a strong assumption, and there exist many DSGE models where key shocks are not partially invertible for usual observables y_t.
- If we only have partial invertibility, cannot identify FVD.
- See Forni, Gambetti & Sala (2019) for details.